Triangulation – II

Strength of Figure

- Strength of figure is a factor to be considered in establishing a triangulation system to maintain the computations within a desired degree of precision
- Very important in deciding the layout of a triangulation system

- US Coast and Geodetic Surveys has developed a convenient method of evaluating the strength of a triangulation figure
- For the square of the probable error (L^2) that would occur in the sixth place of the logarithm of any side

$$L^2 = \frac{4}{3} d^2 R$$

d is the probable error of an observed direction in seconds of an arc

R is the term which represents the shape of the figure

- R represents the shape of the figure

$$R = \frac{D - C}{D} \Sigma(\delta_A^2 + \delta_A \delta_B + \delta_B^2)$$

D = number of directions observed excluding the known side of the figure

δ_A, δ_B = difference per second in the sixth place of logarithm of the sine of the distance angles A and B respectively

$$C = (n' - S' + 1) + (n - 2S + 3)$$

$C =$ number of geometric conditions for side and angle to be satisfied in each figure

$n =$ total number of lines including the known side in a figure

$n' =$ number of lines observed in both directions including the known side

$S =$ total number of stations

$S' =$ number of stations occupied

Exercise

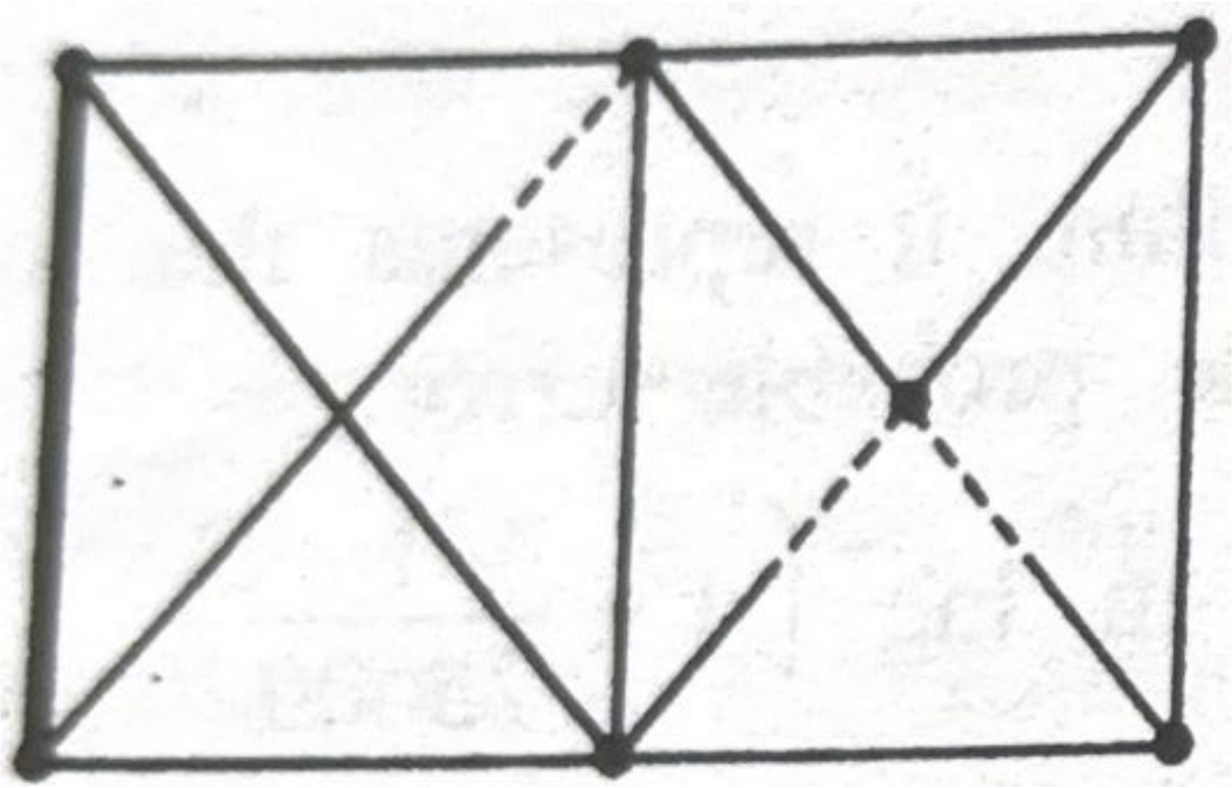
- If the probable error of direction measurement is 1.2”, compute the maximum value of R for the desired maximum probable error of 1 in 20000

L = the 6th place in $\log\left(1 \pm \frac{1}{20000}\right) = \pm 21$
$d = 1.20$“

$$L^2 = \frac{4}{3} d^2 R$$

$$R = \frac{3}{4} . \frac{L^2}{d^2} = \frac{3}{4} . \frac{(21)^2}{(1.2)^2} = 230$$

- Compute the value of $\frac{D-C}{D}$ for the following triangulation figure



$n =$ total number of lines including the known side in a figure

$= 13$

$n' =$ number of lines observed in both directions including the known side

$= 10$

$S =$ total number of stations

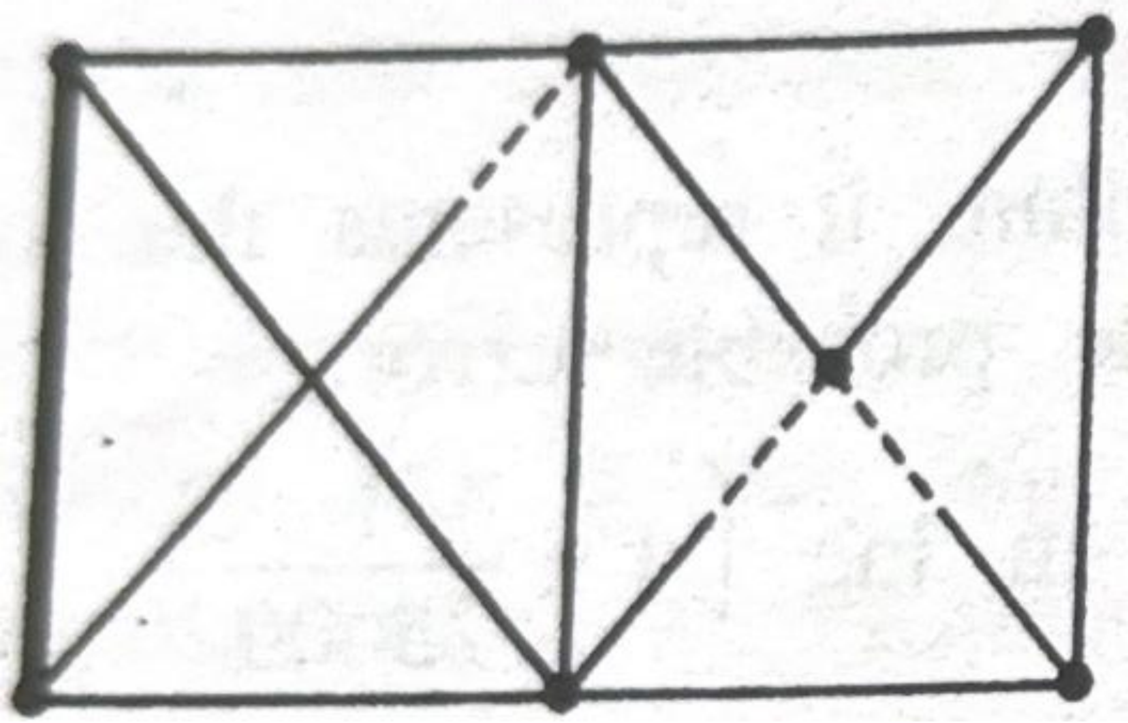
$= 7$

$S' =$ number of stations occupied

$= 7$

$D =$ Total directions observed excluding the known side

$= 21$

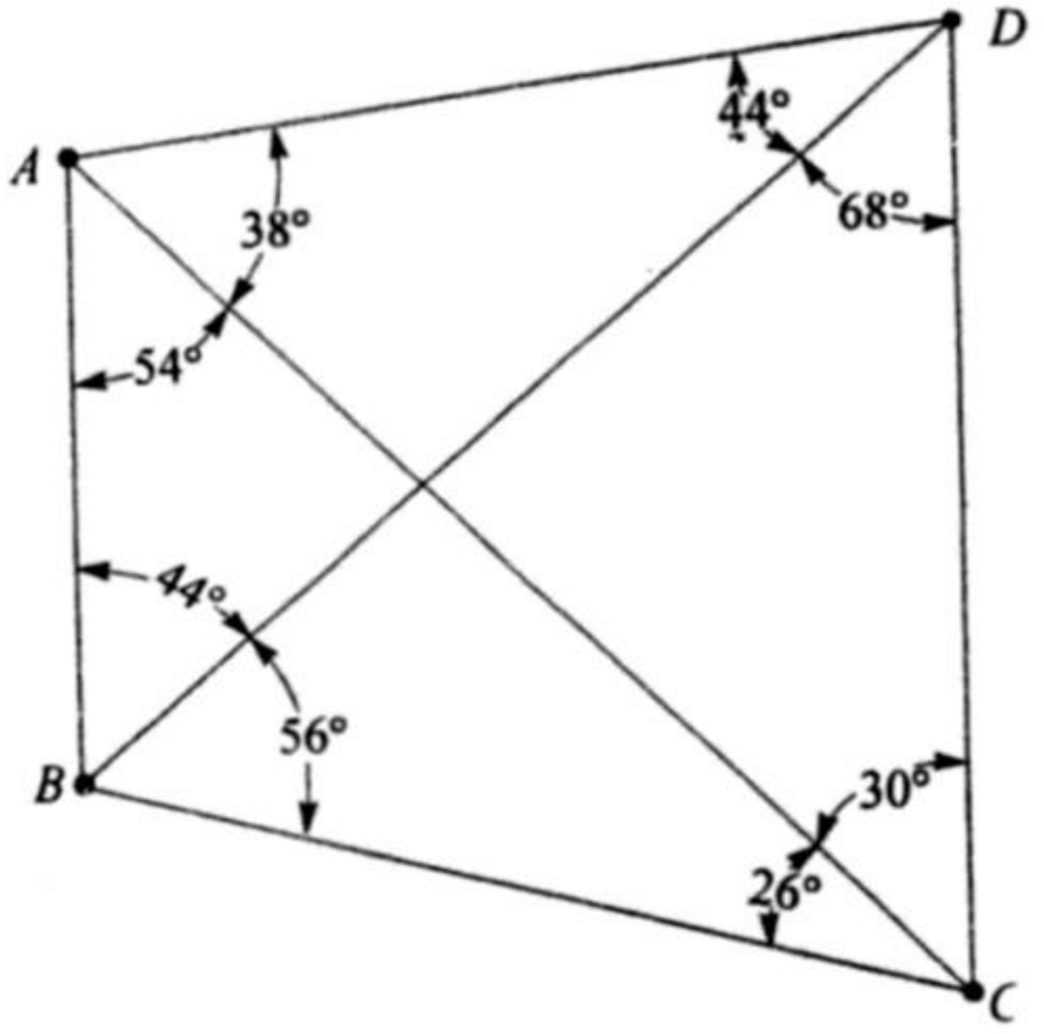


$$C = (n' - S' + 1) + (n - 2S + 3)$$

$$C = (10 - 7 + 1) + (13 - 2 \times 7 + 3) = 6$$

$$\frac{D - C}{D} = \frac{21 - 6}{21} = 0.714$$

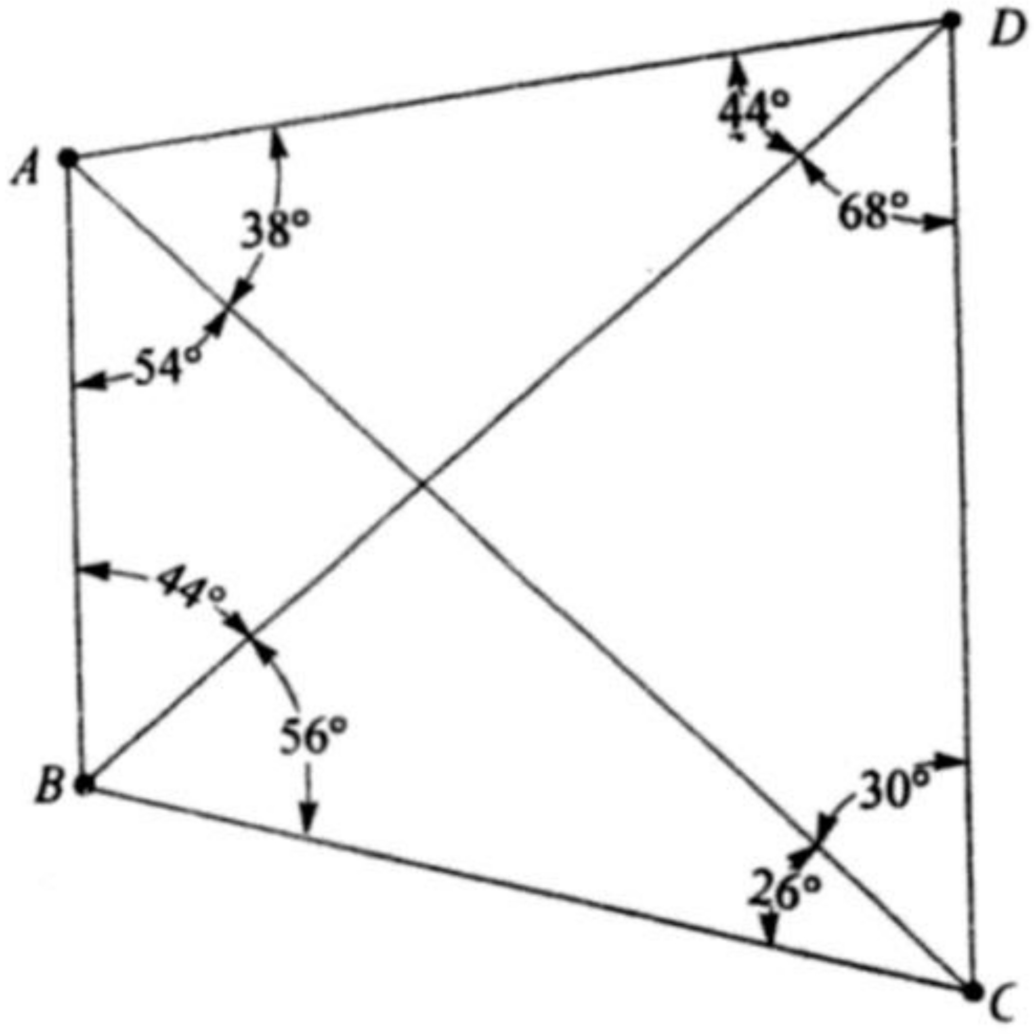
- Compute the strength of the figure $ABCD$ for all the routes by which the length CD can be computed from the known side AB. Assume that all the stations were occupied.



- For the given figure

$$n = 6$$
$$n' = 6$$
$$S = 4$$
$$S' = 4$$
$$D = 10$$
$$C = (6 - 4 + 1) + (6 - 2 \times 4 + 3) = 4$$

$$\frac{D - C}{D} = \frac{10 - 4}{10} = 0.6$$



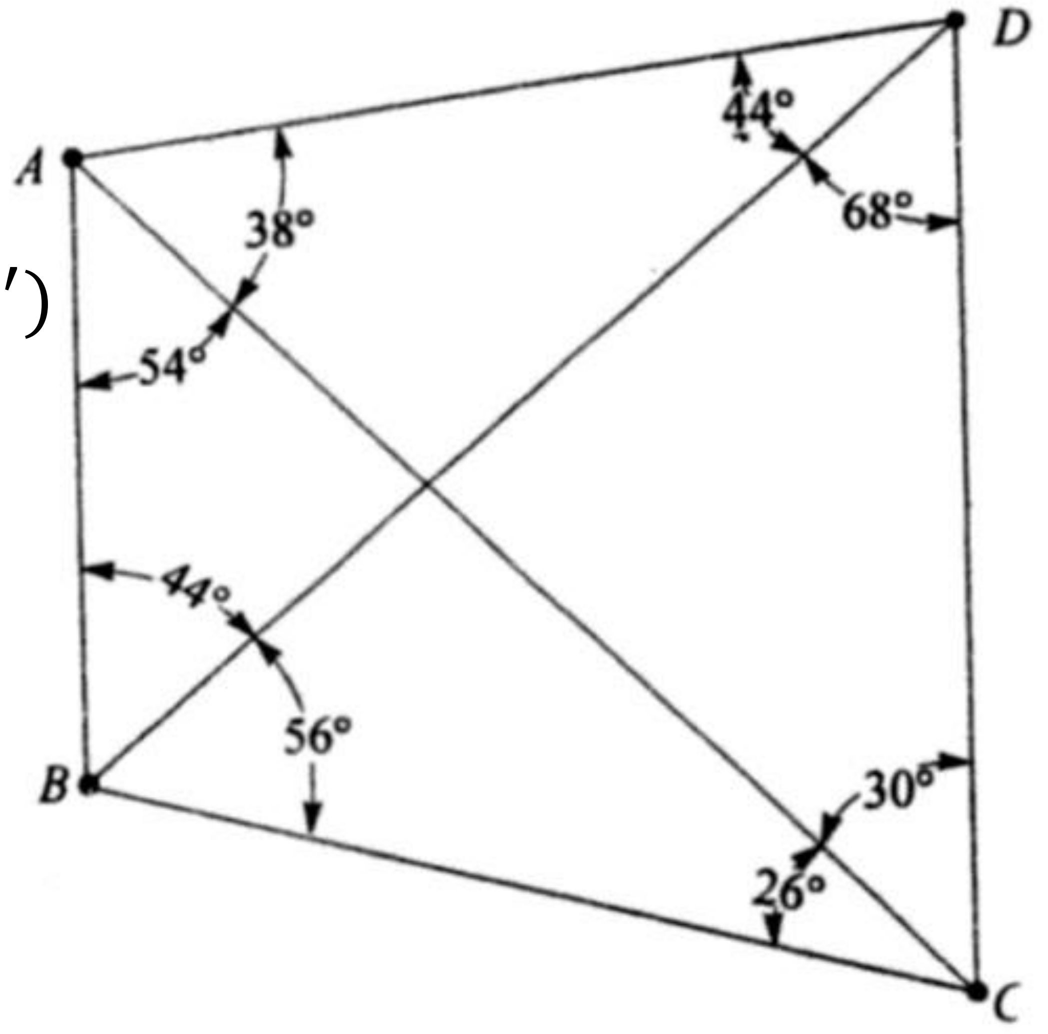
Route 1 (Using ΔABC and ΔADC, common side AC)

For ΔABC, Distance angles are 26° and 100°

$$\delta_{100} = \log\sin(100°0'1'') - \log\sin(100°0'0'')$$
$$= 17$$

$$\delta_{26} = \log\sin(26°0'1'') - \log\sin(26°0'0'')$$
$$= -0.37$$

$$\delta_{100}^2 + \delta_{100}\delta_{26} + \delta_{26}^2 = 17$$



Route 1 (Using ΔABC and ΔADC, common side AC)

For ΔADC, Distance angles are $112°$ and $38°$

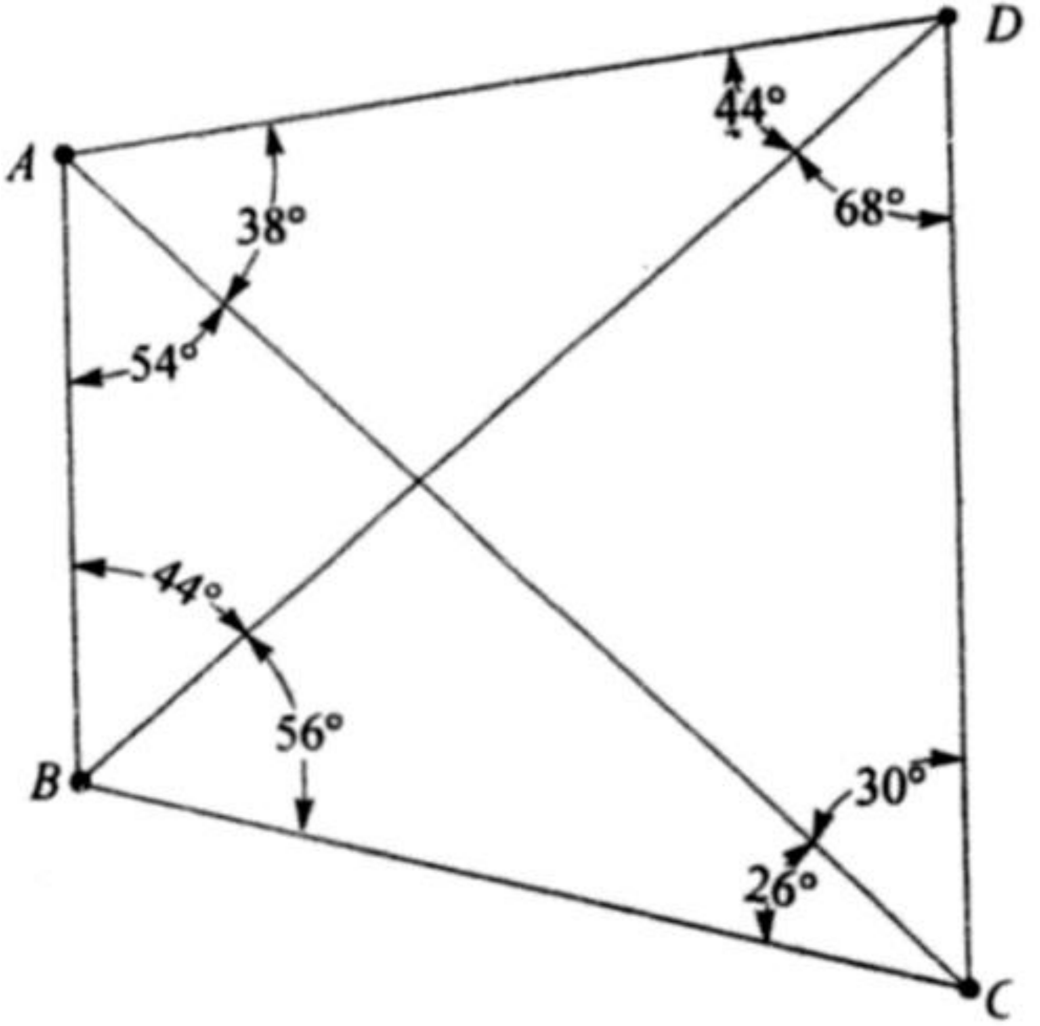
$$\delta_{112} = \log\sin(112°0'1'') - \log\sin(112°0'0'')$$
$$= -0.85$$

$$\delta_{38} = \log\sin(38°0'1'') - \log\sin(38°0'0'')$$
$$= 2.69$$

$$\delta_{112}^2 + \delta_{112}\delta_{38} + \delta_{38}^2 = 6$$

Strength of figure for Route 1

$$R_1 = 0.6 \times (17 + 6) = 14$$



Route 2 (Using ΔABC and ΔBCD, common side BC)

For ΔABC, Distance angles are 26° and 54°

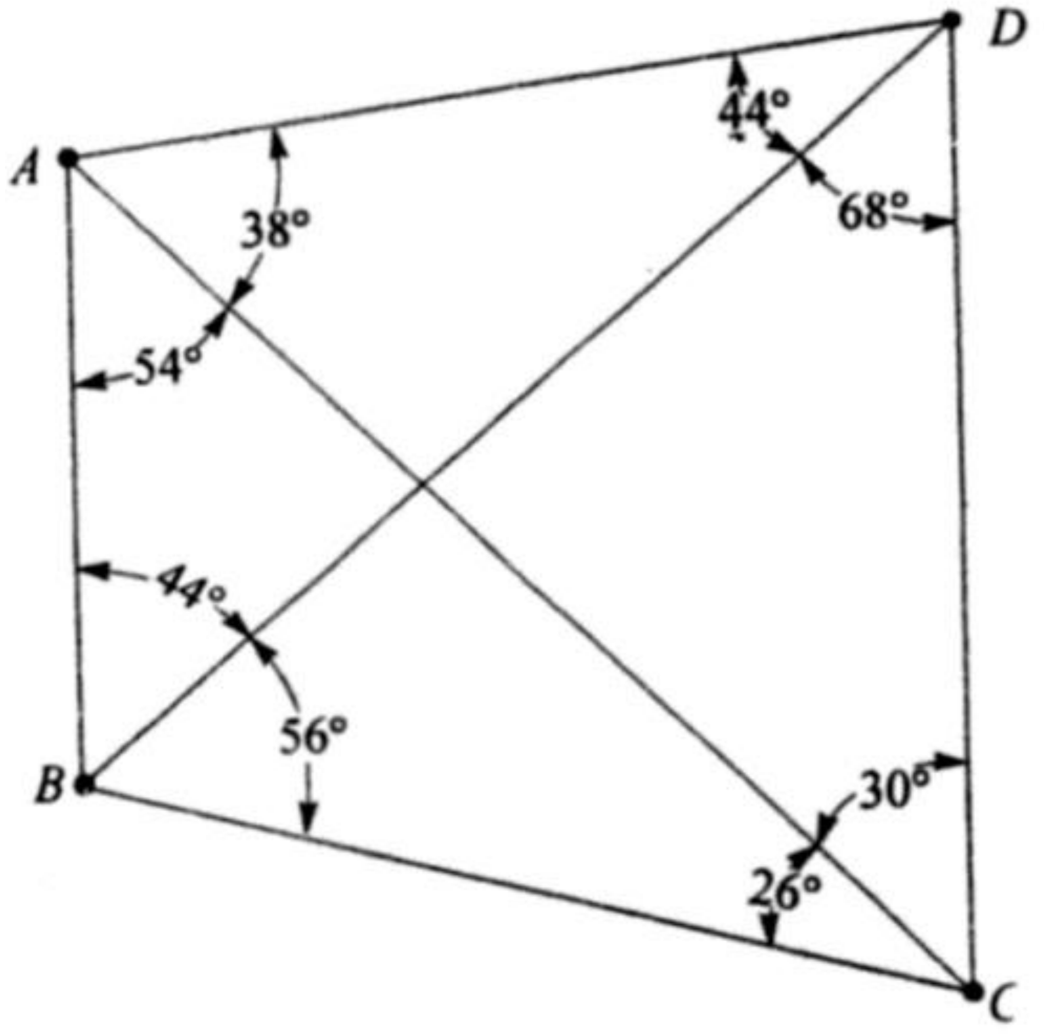
$$\delta_{54}^2 + \delta_{54}\delta_{26} + \delta_{26}^2 = 27$$

For ΔBCD, Distance angles are 68° and 56°

$$\delta_{56}^2 + \delta_{56}\delta_{68} + \delta_{68}^2 = 4$$

Strength of figure for Route 2

$$R_2 = 0.6 \times (27 + 4) = 19$$



Route 3 (Using ΔABD and ΔACD, common side AD)

For ΔABD, Distance angles are 44° and 44°

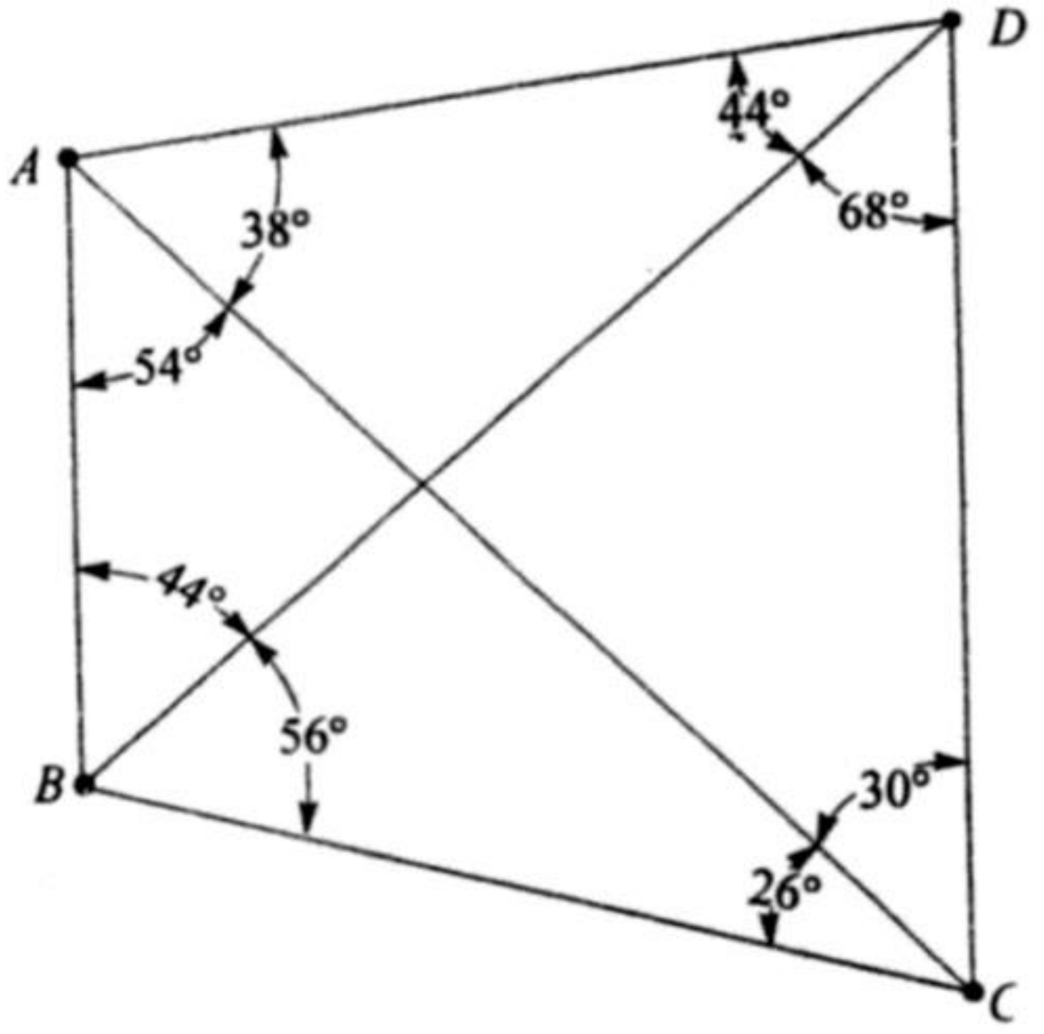
$$\delta_{44}^2 + \delta_{44}\delta_{44} + \delta_{44}^2 = 13$$

For ΔACD, Distance angles are 30° and 38°

$$\delta_{38}^2 + \delta_{30}\delta_{38} + \delta_{30}^2 = 31$$

Strength of figure for Route 3

$$R_3 = 0.6 \times (13 + 31) = 26$$



Route 4 (Using ΔABD and ΔBCD, common side BD)

For ΔABD, Distance angles are 44° and 92°

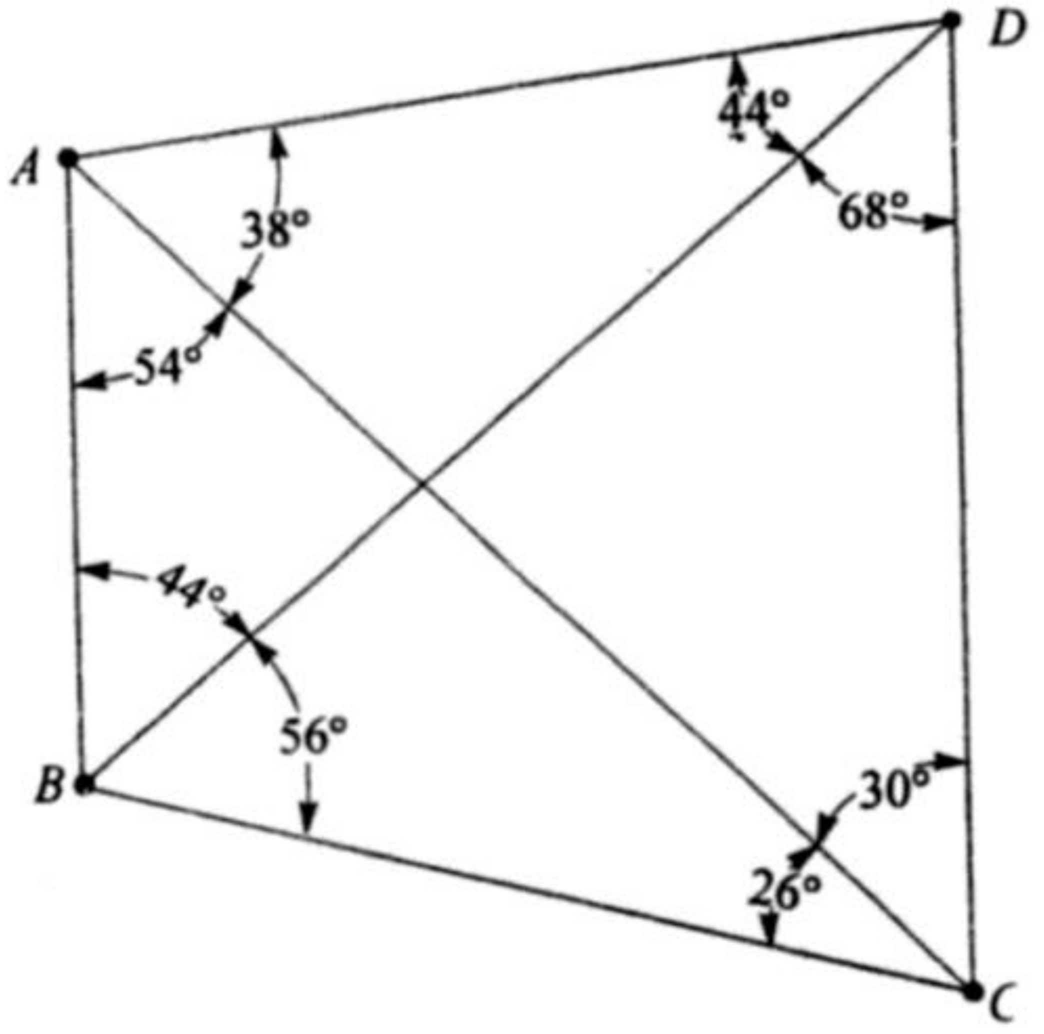
$$\delta_{92}^2 + \delta_{44}\delta_{92} + \delta_{44}^2 = 7$$

For ΔBCD, Distance angles are 56° and 56°

$$\delta_{56}^2 + \delta_{56}\delta_{56} + \delta_{56}^2 = 7$$

Strength of figure for Route 4

$$R_4 = 0.6 \times (7 + 7) = 8$$



Since the lowest value of R represents the highest strength, the best route to compute the length of CD is Route 4, having $R_4 = 8$

Maximum value of R recommended by US Coast and Geodetic Surveys

	First Order		Second Order		Third Order	
	R_1	R_2	R_1	R_2	R_1	R_2
Single Independent figure						
Desirable	15		25	80	25	120
Maximum	25	80	40	120	50	150
Net between bases						
Desirable	80		100		125	
Maximum	110		130		175	

Satellite Station and Reduction to Centre

- In order to secure well conditioned triangle or better visibility, objects such as towers, chimneys, flag poles, spires etc. are sometimes selected as triangulation stations
- It is impossible to set up an instrument over the raised stations for taking observations
- Hence, a subsidiary station is selected near the main station, and observations are taken to the other triangulation stations with the same precision
- These stations are known as Satellite Station, or Eccentric Station or False Station

- These angles are later corrected and reduced to what they would have been if the true station was occupied
- The operation of applying the corrections due to the eccentric of the station is known as ‘Reduction to Centre’

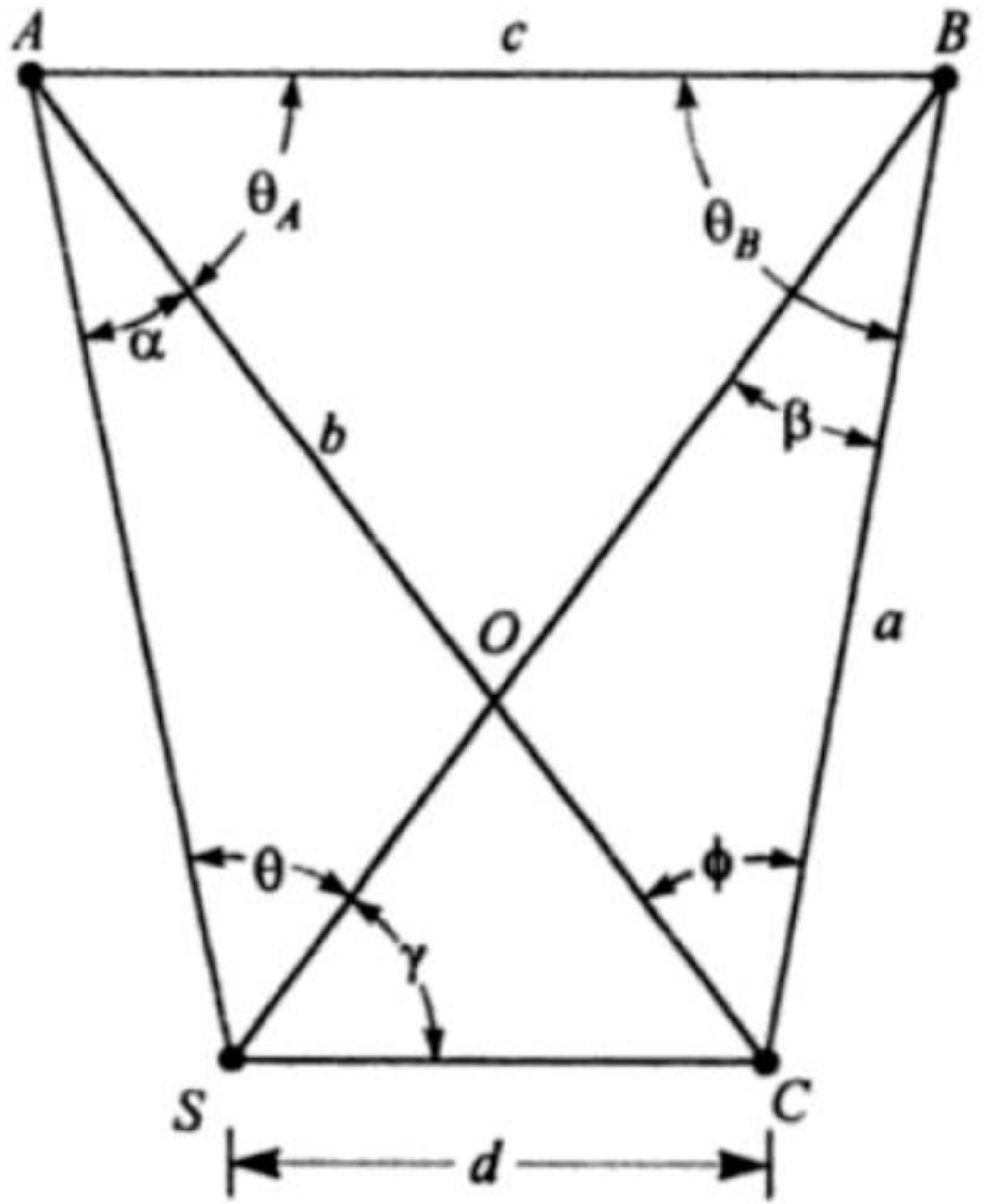
- A, B, C are triangulation stations.
- Observations from C are not possible, the observations from S are made on A, B and C
- From ΔABC

$$\frac{c}{sin\phi} = \frac{a}{sin\theta_A} = \frac{b}{sin\theta_B}$$

$$b = \frac{c.\,sin\theta_B}{sin\phi}$$

- From ΔSAC

$$\frac{d}{sin\alpha} = \frac{b}{sin(\theta + \gamma)}; sin\alpha = \frac{dsin(\theta + \gamma)}{b}$$

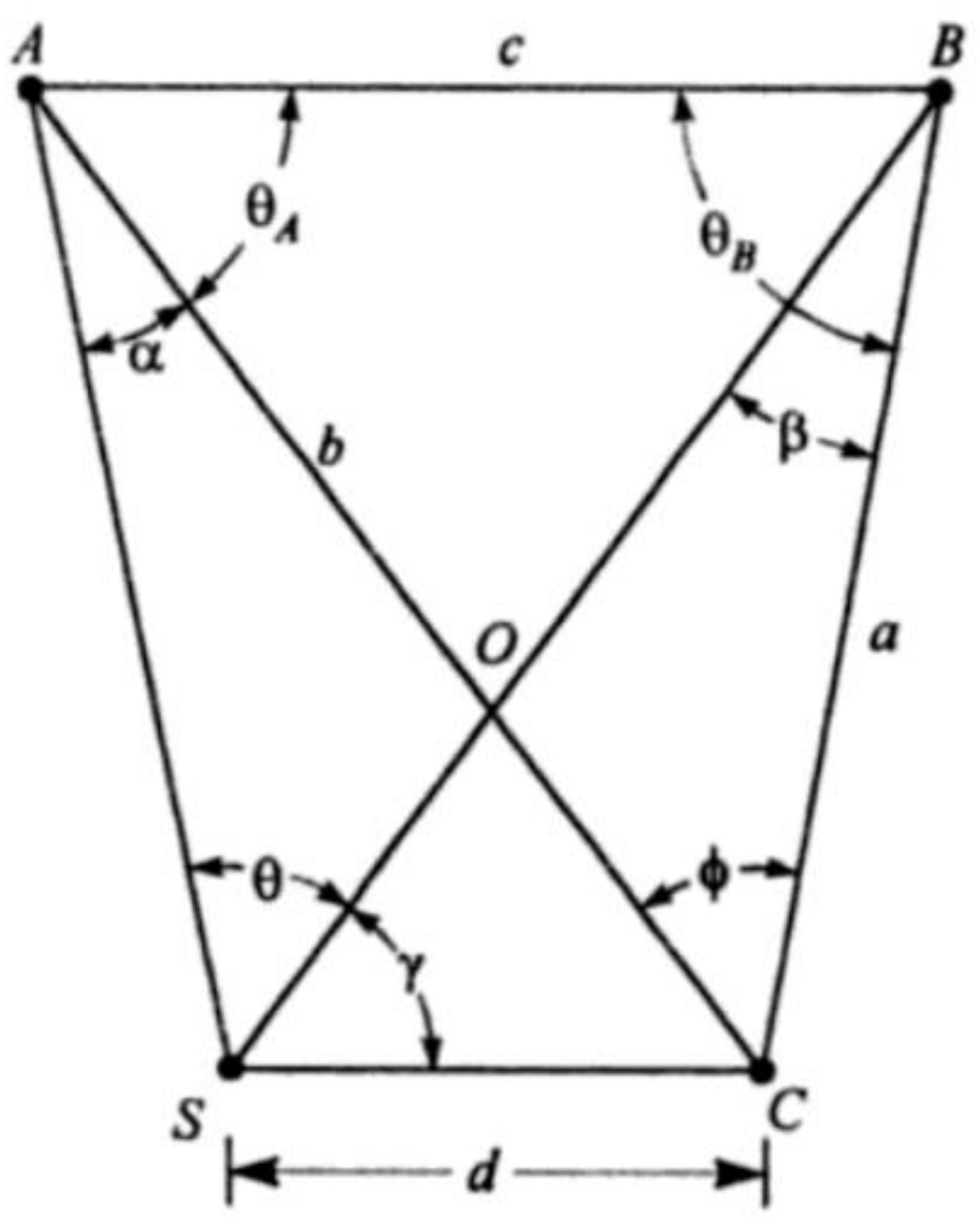


- From ΔSBC

$$\frac{d}{sin\beta} = \frac{a}{sin\gamma}; sin\beta = \frac{d.\,sin\gamma}{a}$$

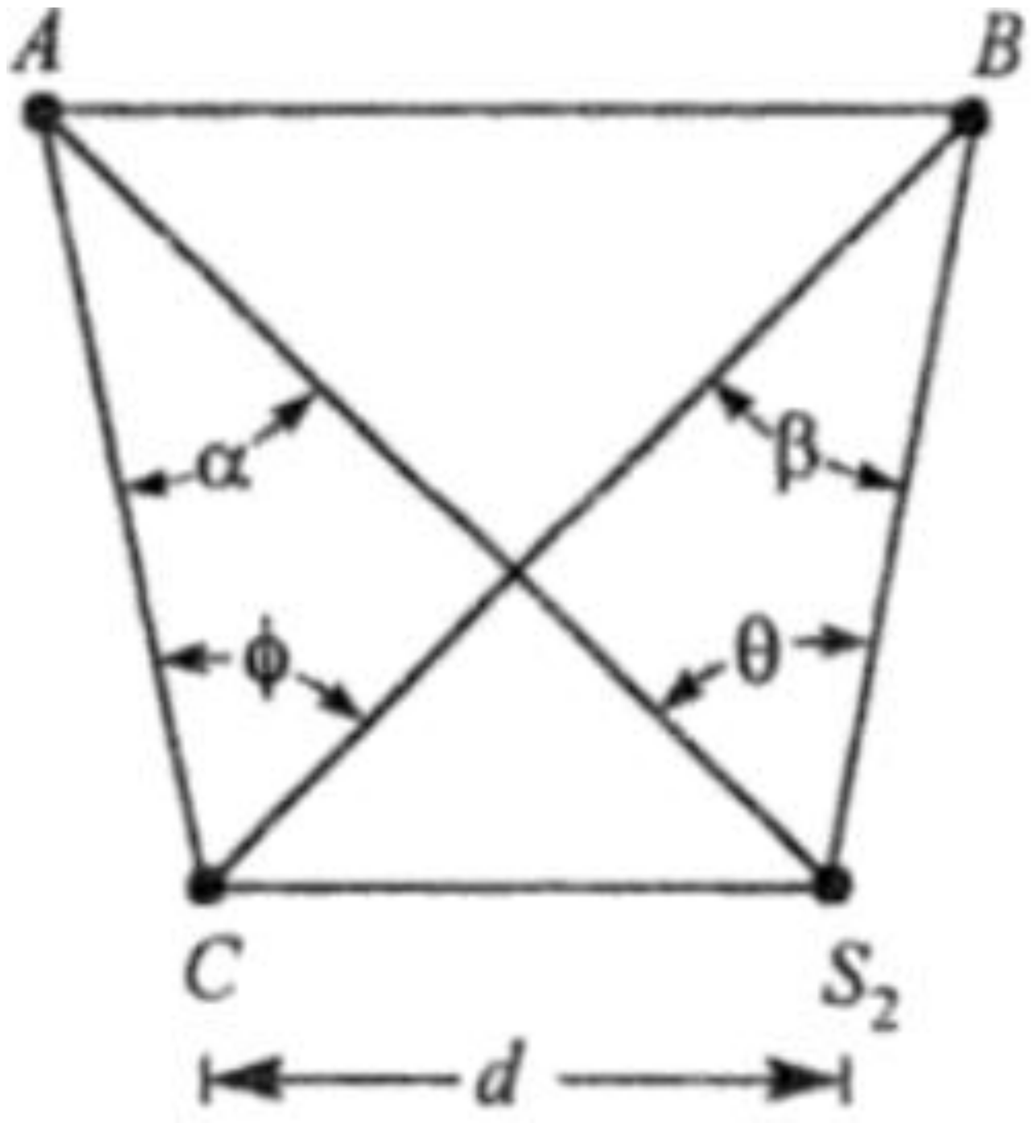
- *Case I: S towards left of C*

$$\phi = \theta + \alpha - \beta$$



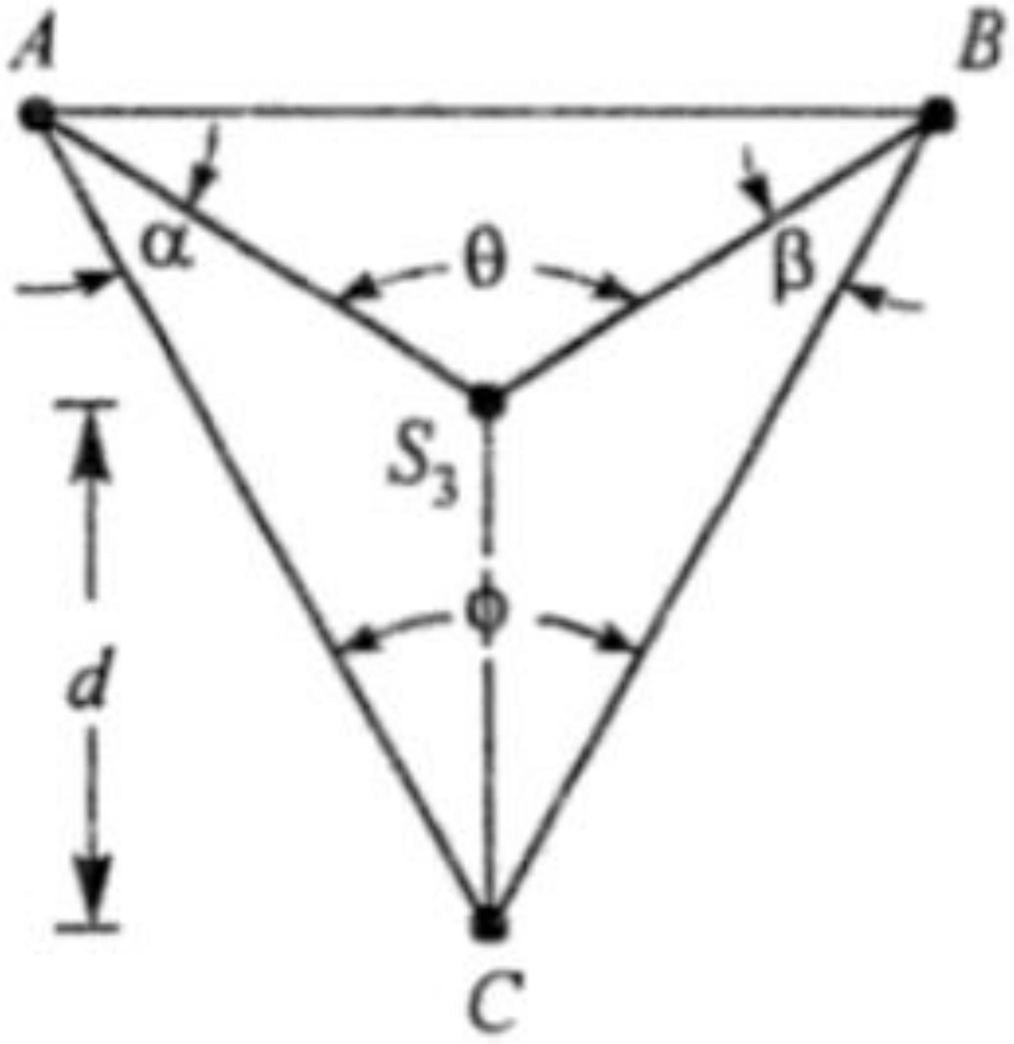
- *Case II: S towards right of C*

$$\phi = \theta - \alpha + \beta$$



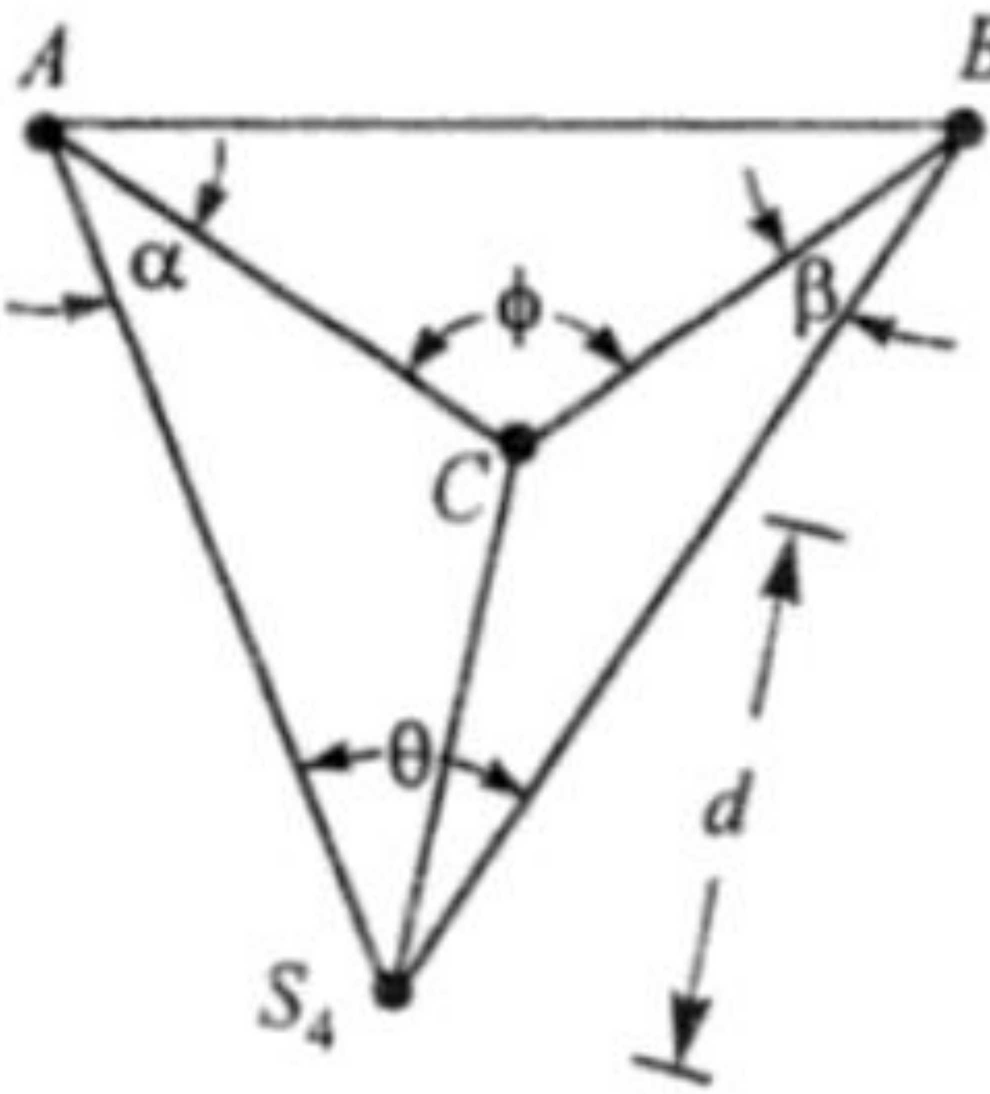
- *Case III: S inside* ΔABC

$$\phi = \theta - \alpha - \beta$$



- *Case III: S outside* ΔABC

$$\phi = \theta + \alpha + \beta$$



Exercise

- Directions were observed from a satellite station S, $80m$ from C with following results. $A(0°00'00'')$, $B(72°50'44'')$, $C(299°22'00'')$. The approximate lengths AC andBC are $17km$ and $24.15km$, respectively. Compute the angle ACB

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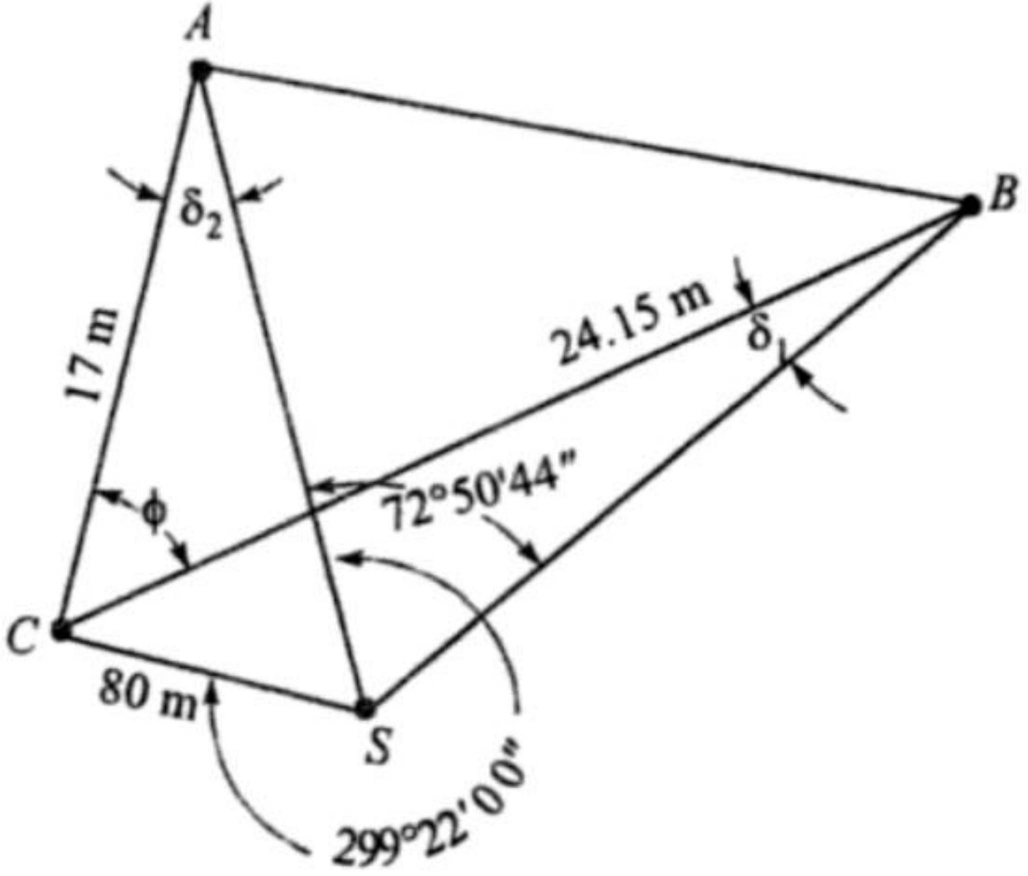

- From ΔSBC

$$\frac{80}{sin\delta_1} = \frac{24.15 \times 1000}{sin[(72°50'44'' + (360 - 299°22'00'')]}$$

$$\delta_1 = 0°8'15.81''$$

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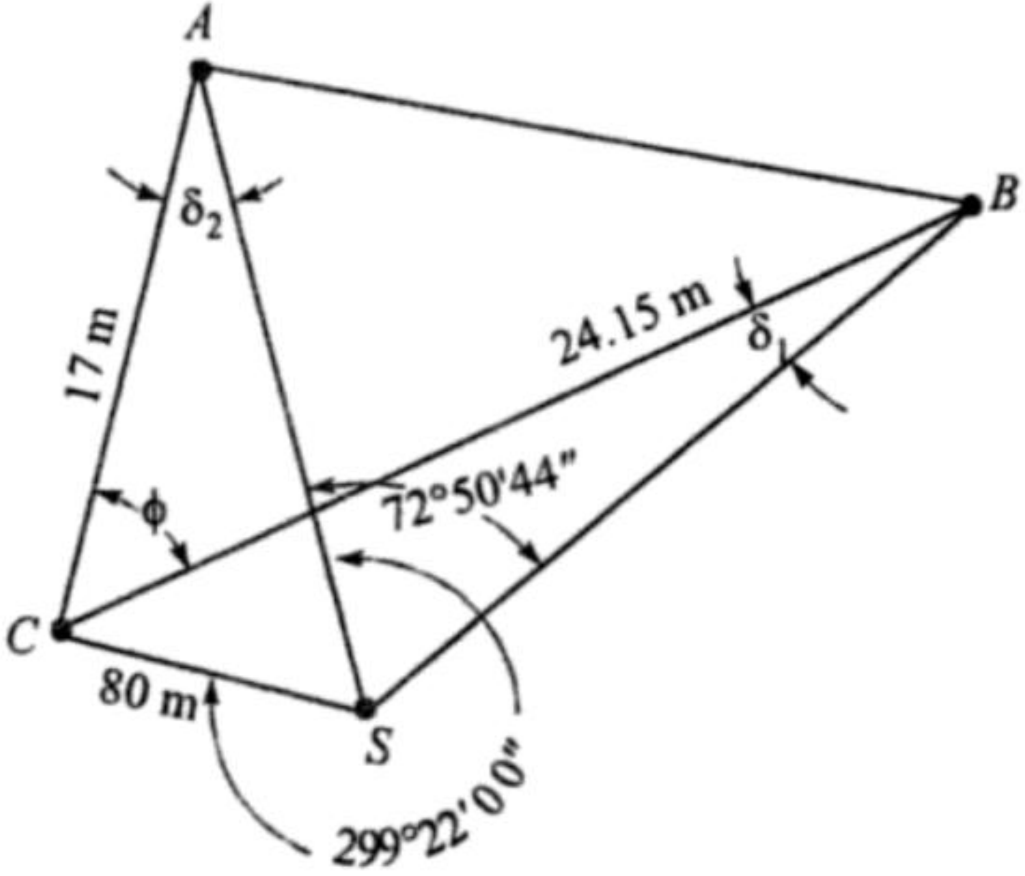
- From ΔSAC

$$\frac{80}{sin\delta_2} = \frac{17 \times 1000}{\sin(360 - 299°22'00'')}$$

$$\delta_2 = 0°14'\,5.93''$$

$$\phi = 72°50'44'' - 0°14'5.93'' + 0°8'15.81''$$



Exercise

- A, B and C are the stations in a minor triangulation survey. A satellite station S is set up near C such that AC and BC fall within triangle ASB. It is given that $AC = 7.5km$, $BC = 6.3km$, $CS = 40.5m$. $\angle ASC = 62°10'40''$ and $\angle ASB = 75°15'15''$. Calculate $\angle ACB$

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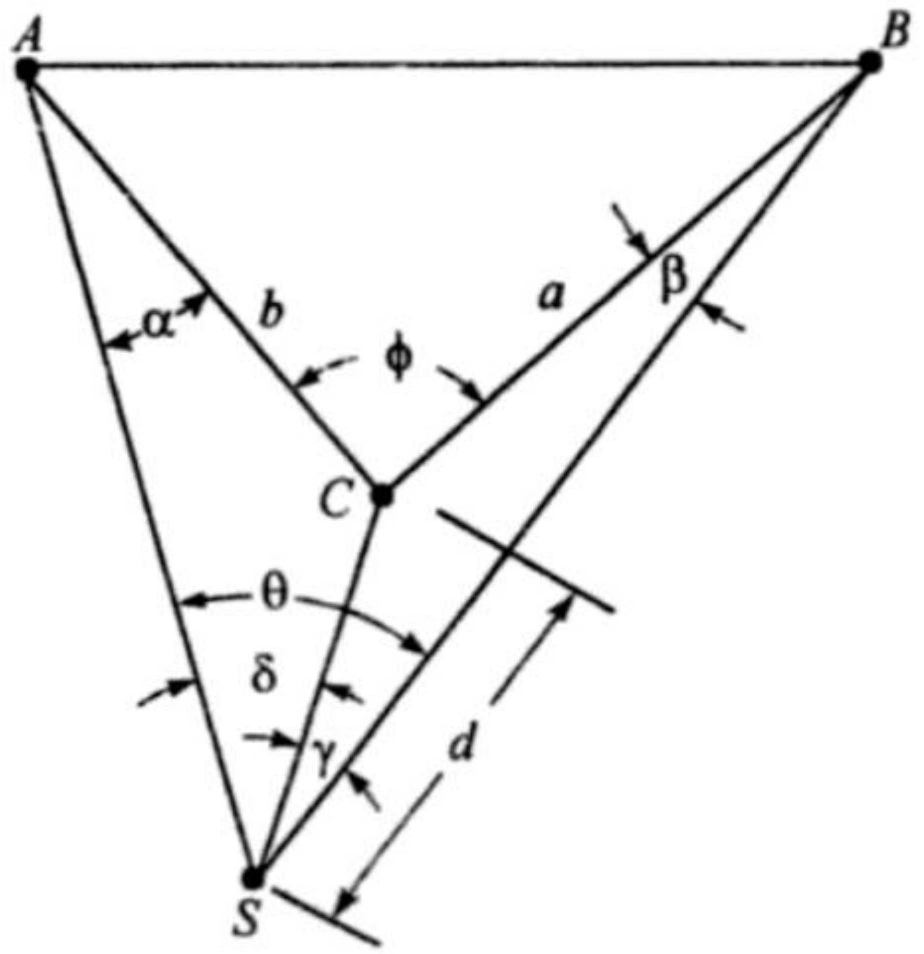
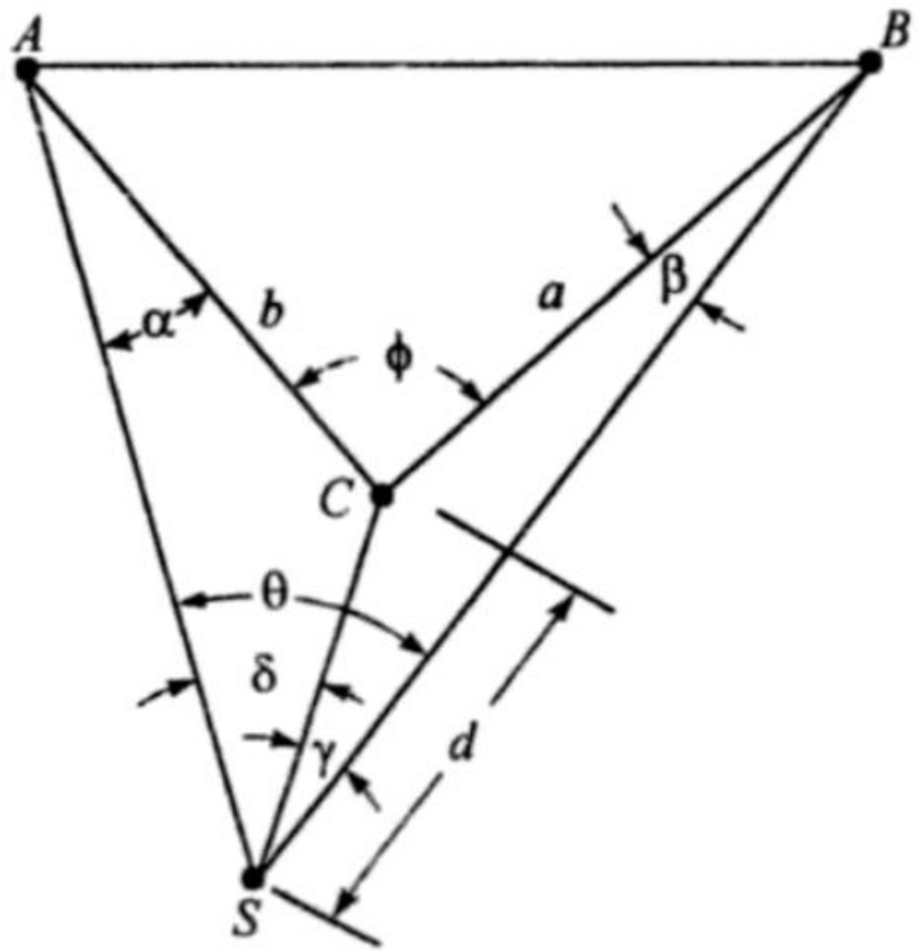


- From ΔACS

$$\frac{CS}{sin\alpha} = \frac{AC}{sin\delta}; sin\alpha = \frac{CS.sin\delta}{AC}$$

$$sin\alpha = \frac{40.5\ sin(62°10'40'')}{7.5 \times 1000}$$

$$\alpha = 0°16'25.07''$$



- From ΔBCS

$$\frac{CS}{sin\beta} = \frac{BC}{sin\gamma}; sin\beta = \frac{CS.sin\gamma}{BC}$$

$$\gamma = (75°15'15'' - 62°10'40'')$$
$$\gamma = 13°04'35''$$

$$sin\beta = \frac{40.5\ sin(13°04'35'')}{6.3 \times 1000}$$

$$\alpha = 0°05'00''$$

$$\phi = \theta + \alpha + \beta$$
$$\phi = 75°15'15'' + 0°16'25.07'' + 0°05'00''$$
$$\phi = 75°36'40.07''$$